

Stat 427: R Programming

Instructor: Benjamin Ridenhour

Fall, 2022

E-mail: bridenhour@uidaho.edu

Office Hours: TR 3:00–4:00pm

Office: Brink Hall 327

Web: <https://bit.ly/3C33Hyr>

Class Hours: TR 9:30–10:45am

Class Room: TLC 145

Course Description

In this course, we will focus on maximizing your ability to use the R language and some of its associated helper programs (namely RStudio, R Markdown, and R Shiny). R has become one of the dominant scientific programming languages due to being a freely-available, open-source language with a large library of graphical, statistical, and mathematical functions. When working on statistical or mathematical models, we often find that currently available routines do not cover your needs as a researcher. You must be able to modify existing routines or create your own programs *de novo* when this situation arises. When errors arise in either your programs or those of others, you must be able to successfully troubleshoot the problem. In this course you will learn how to create your own R programs, read and interpret R documentation, use existing R packages, troubleshoot errors, and optimize R code. Because there is no formal programming prerequisite for this course, we will also cover some basic programming concepts shared by many programming languages (e.g., Python, C++).

Course Structure

Our class is intended to be an in-person class, and, as such, content delivery will be done via a live lecture. Attendance is particularly important in this class as we will spend part of each class period doing hands-on practicum using your own computer. Obviously, given the circumstances of the ongoing COVID-19 pandemic, students will not always be able to attend class; I will do my best to accommodate situations as they arise. I highly recommend you attend if at all possible. Similarly, evaluation (exams, etc) are being done in-person (see below).

Coursework will be as follows. Each period, I will present key concepts from the chapter of the book being covered. As a class, we will go through the code blocks in each chapter and work on the exercises. At the end of each class period, you will be asked to turn in an RMD file with your work. Doing the in-class work will be 25% of your overall grade. The graded portion of the class will be your performance on three equally-weighted, *cumulative* exams. By cumulative, I do not necessarily mean that questions on each exam will come from earlier chapters, but that the

knowledge presented in the class cumulative and it is hard to ask about later topics without know prior topics. Each exam will also be worth 25% of your grade. **If you cannot be in attendance for an exam, arrangements need to be made with me prior to the event.**

Required Materials

- *Advanced R, Second Edition*. By Hadley Wickham. CRC Press: Boca Raton, USA.
- You will need a laptop that has the following software installed on it: R (available at [CRAN](#)) and RStudio (available [here](#)).

Prerequisites/Corequisites

Prerequisites: Either Stat 251, 301, or 416

Topics By Week (*Approximate*)

1. Wk 1: Introduction; Ch. 2 Names and Values
2. Wk 2: Ch. 3 Vectors; Ch. 4 Subsetting
3. Wk 3: Ch. 5 Flow control; Ch. 6 Functions
4. Wk 4: Ch. 7 Environments; Ch. 8 Conditions
5. Wk 5: Exam 1; Ch. 9 Functionals
6. Wk 6: Ch. 10 Function factories; Ch. 11 Function operators
7. Wk 7: Ch. 12 Object oriented programming; Ch 13. S3
8. Wk 8: Ch. 14 R6; Ch 15. S4
9. Wk 9: Ch. 17 Metaprogramming; Ch. 18 Expressions
10. Wk 10: Exam 2; Ch. 19 Quasiquote
11. Wk 11: Ch. 20 Evaluation; Ch 22. Debugging
12. Wk 12: Ch. 23 Performance; Ch. 24 Improving performance
13. Wk 13: Ch. 25 R in C++; tidyverse
14. Wk 14: *Fall Break 21–25 Nov.*
15. Wk 15: R Shiny; reticulate
16. Wk 16: Dead week/Review

Course Objectives/Learning Outcomes

Successful students will:

1. Understand how to program, use available packages, and troubleshoot errors in R.

Final Exam Date

16 Dec (Friday), 8:00–10:00am

Grading Policy

A typical grading scale will be used with cutoffs every 10% (i.e., $\geq 90\%$ = A, $(90\%, 80\%]$ = B, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 25% of your grade will be determined by each of the 3 exams.
- 25% of your grade will be determined by your in-class work.